

euclid

proof_machine

Verified by the sealed kernel V

Statement

Theorem 1 (euclid).

$$\forall n:\mathbb{N}. \exists p:\mathbb{N}. ((p > n) \wedge \text{prime}(p))$$

Declared axioms

The proof cites the following theory-package axioms (trusted assumptions for v0.1):

- **fact_pos**: $\forall n:\mathbb{N}. ((n! + 1) \geq 1)$
- **has_prime_divisor**: $\forall m:\mathbb{N}. \exists p:\mathbb{N}. (\text{prime}(p) \wedge (p \mid m))$
- **euclid_step**: $\forall p:\mathbb{N}. \forall n:\mathbb{N}. ((\text{prime}(p) \wedge (p \mid (n! + 1))) \Rightarrow (p > n))$

Proof trace

Each row is one primitive kernel step. The trace was verified by **vrfy** as a Cook–Reckhow certificate.

#	rule	conclusion	from
1	axiom [has_prime_divisor]	$\forall m:\mathbb{N}. \exists p:\mathbb{N}. (\text{prime}(p) \wedge (p \mid m))$	–
2	$\forall$ -elim (SPEC) at $(n! + 1)$	$\exists p:\mathbb{N}. (\text{prime}(p) \wedge (p \mid (n! + 1)))$	1
3	assume at $(\text{prime}(p) \wedge (p \mid (n! + 1)))$	$(\text{prime}(p) \wedge (p \mid (n! + 1)))$	–
4	$\wedge$ -elim left	$\text{prime}(p)$	3
5	axiom [euclid_step]	$\forall p:\mathbb{N}. \forall n:\mathbb{N}. ((\text{prime}(p) \wedge (p \mid (n! + 1))) \Rightarrow (p > n))$	–
6	$\forall$ -elim (SPEC) at p	$\forall n:\mathbb{N}. ((\text{prime}(p) \wedge (p \mid (n! + 1))) \Rightarrow (p > n))$	5
7	$\forall$ -elim (SPEC) at n	$((\text{prime}(p) \wedge (p \mid (n! + 1))) \Rightarrow (p > n))$	6
8	modus ponens	$(p > n)$	7, 3
9	$\wedge$ -intro	$((p > n) \wedge \text{prime}(p))$	8, 4
10	$\exists$ -intro (EXISTS) binding q at p	$\exists q:\mathbb{N}. ((q > n) \wedge \text{prime}(q))$	9
11	$\exists$ -elim (CHOOSE) with $p:\mathbb{N}$	$\exists q:\mathbb{N}. ((q > n) \wedge \text{prime}(q))$	2, 10
12	$\forall$ -intro (GEN) with $n:\mathbb{N}$	$\forall n:\mathbb{N}. \exists q:\mathbb{N}. ((q > n) \wedge \text{prime}(q))$	11

Provenance

The trusted base for this document is the kernel verifier V (see `kernel/verify.ml`) plus the axioms listed above.